

A Short Report on Simulation-Based Inference

Scientific Machine Learning Seminar

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Abstract

Simulation-Based Inference (SBI) offers a novel approach to solve inverse problems powered by machine learning. Hereby, a neural network learns to map observed data to a posterior distribution over model parameters. By using simulations, SBI does not require any likelihood evaluation which otherwise often is computationally costly or has to be approximated. This can allow SBI to outperform classical Bayesian inference methods in inference speed and posterior accuracy. However, common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the usefulness of SBI if they are not circumvented. This report presents basic SBI methodology, in particular Neural Posterior Estimation, discusses its advantages and limitations and reviews two exemplary applications for SBI: detecting planetary events from gravitational microlensing and inferring black hole merger parameters from gravitational waves.

1 Introduction

In computational physics there are usually two types of problems to solve: Firstly, forward problems where the parameters and/or boundary conditions of a model are given and the goal is to find data or a data distribution which adheres to this model. For example in the case of given parameters and initial values this is a typical simulation.

Though secondly, there are also inverse problems where the data is the preset and now the parameters or parameter distributions of the assumed model have to be inferred. So here the data likely comes from an observation and the parameters which represent physical properties of the system have to be computed. This is the type of problem that will be explored in this report.

A rather simple case of solving an inverse problem is the use of linear regression to fit a function to a given dataset. But for more complex models also methods with higher efficiency are needed. The classical approach here is to use Bayesian methods to infer the parameter posterior distribution, but Simulation-Based Inference introduces machine learning techniques to solving inverse problems. SBI is able to solve these problems in some cases more efficiently and computationally cheaper. Applications where SBI shows advantages are (but not limited to) gravitational microlensing (Zhang et al., 2021), gravitational wave physics (Dax et al., 2023) and neuroscience (Deistler et al., 2025, Sec. 4.2).

In Section 2 the general concepts of classical Bayesian inference are first explored, specifically focussing on Markov-Chain-Monte-Carlo methods. Then SBI and its underlying concepts are

introduced in Section 3 and exemplified in Section 4 with the help of a toy-example. The advantages and limitations of SBI are presented in Section 5. Two exemplary applications with results are portrayed in Section 6 and a conclusion is given in Section 7.

2 Classical Bayesian Inference

The most outstanding feature of Bayesian Inference in comparison to regression is that the output of BI is a posterior distribution of the parameters instead of just the most likely parameter set sampled from this distribution. So instead of just computing $\theta \sim p(\theta|x)$, the result is the full $p(\theta|x)$ distribution. The naming convention here is that θ is the set of parameters from the model and x is the observed data.

The underlying concept is the Bayes equation

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \quad (1)$$

from Bayesian statistics where $p(\theta|x)$ is the posterior distribution, $p(x|\theta)$ the likelihood, $p(x)$ the evidence and $p(\theta)$ the prior distribution. While the prior can be chosen adequately according to assumed distributions of the model parameters and the evidence is just a normalizing factor, the likelihood $p(x|\theta)$ has to somehow be calculated, evaluated or approximated in numerical methods of inferencing the posterior. This is the most important and also most costly step of classical Bayesian Inference (Deistler et al., 2025).

Different methods of Bayesian inference solve this in different ways. Markov chain Monte Carlo (MCMC) methods are relatively simple and widely used for inferring posteriors. These evaluate the likelihood many times randomly in the parameters space and build a posterior distribution from that. In Metropolis-Hastings, a MCMC method, the likelihood value at a random point in the parameter space determines if this point is accepted or rejected. The accumulated distribution of the accepted points then converges to the posterior after many evaluations (Prado et al., 2026).

There are adjusted methods that can make the algorithm more efficient (like introducing random walk instead of true random sampling) but all classical Bayesian inference methods suffer from similar key limitations: The convergence to a posterior that is accurate enough for the specific application takes many evaluations of the likelihood and is therefore very computationally expensive. For the applications that are later shown in Section 6 this can be up to 1 second per evaluation, for an adequately sampled posterior distribution the computation time per analyzed dataset can be huge.

In complex models the exact evaluation of the likelihood is often not possible or a lot more computationally expensive, so it has to be approximated by a Gaussian distribution, usually as part of the χ^2 method. This inserts another error source into the posterior distribution.

Classical Bayesian inference methods also have the problem that they cannot model multimodal posteriors very well. The algorithm often gets stuck in one mode and is unable to jump to another, this becomes a problem in models with a lot of close degeneracies (Deistler et al., 2025). In turn, grid searches and human intervention are often needed increases the time and effort of the data analysis process. This may not be a possibility for the huge amounts of data that needs to be processed in modern science.

3 Introduction to Simulation-Based Inference

Key concepts:

Skip Likelihood inference, implicit in simulation

Neural Posterior Estimation (NPE)

Sampling from simulator, train neural network, estimate posterior from observation Fig. 1

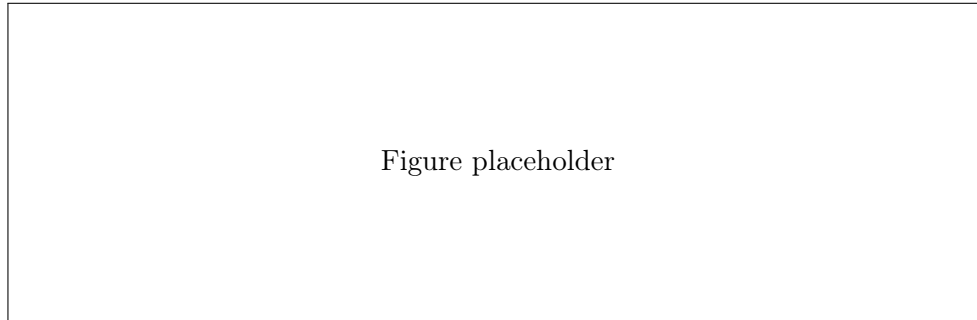


Figure 1: Workflow of SBI.

4 Simple SBI Example

Example: angled throw

choose parameter prior, create simulator of throw, simulate dataset

train neural network, here normalizing flows, any network works

loss function

$$L(\phi) = \tag{2}$$

use network to evaluate observation, data and param interval

5 Benefits and Limitations

Key benefits: very fast inference (Amortized Inference), Likelihood-free (no approximation or expensive computation), multimodal posteriors

Limitations: Upfront training, model misspecification (common pitfall) posterior is a neural estimation uncalibrated posterior

6 Application

Short explanation of microlensing

Results of usage of SBI ([Zhang et al., 2021](#))

Short explanation of gravitational waves and black hole mergers

use of flow matching networks

results of usage here (Dax et al., 2023)

Other applications

7 Conclusion

Benefits and Limitations (stuff from abstract)

When to use SBI inverse problem, time critical inference, expensive likelihood evaluation, multimodal posterior

References

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